Atharva Dangra

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Chennai, Tamil Nadu

EDUCATION

SRM Institute of Technology — B.Tech, Computer Science Engineering *Expected: May 2028*

CGPA: 9.11 (Till 3rd Semester)

The State University of New York — Diploma in Project Management *Feb 2025 – Aug 2025*

Grade: B+

EXPERIENCE

Hackathon Experience *2024 – Present*

- Participated in 10+ hackathons, building real-world solutions under time constraints
- Secured **4th place** in Hackelite 2026 (National Level Hackathon, SRMIST Vadapalani)
- Developed multiple project prototypes across AI, web, and simulation domains
- Collaborated in team environments, contributing to both development and project pitching

PROJECTS

MatterGen – AI Material Property Predictor | *Python, Streamlit, scikit-learn, RDKit*

- Built an ML web app to predict material properties from molecular structures using **Random Forest (scikit-learn)** with RDKit features & fingerprints
- Implemented similarity search using **Tanimoto coefficient** and interactive **3D visualization (Py3Dmol)**
- Developed modular backend for feature extraction and prediction pipeline; deployed with real-time prediction capabilities

MatterGen V2.0 – AI-Assisted Materials Discovery (*In Development*) | *Python, Streamlit, Pandas, NumPy, RDKit, Scikit-learn / PyTorch*

- Developing an AI platform for **material property analysis and discovery** under varying environmental conditions
- Implementing **similarity algorithms** for compound recommendation; integrating RDKit and materials datasets for real-world analysis

GeoTwin AI – Digital Twin for Environmental Simulation | *React, Vite, Mapbox GL JS, Tailwind CSS, JavaScript*

- Built an interactive **digital twin platform** to simulate environmental changes (flood risks, vegetation loss) in real time
- Enabled comparison of present vs. future scenarios through layered **geospatial heatmaps** and mitigation strategy testing

Erylitics – Academic Evaluation & Plagiarism Detection Platform | *Next.js, TypeScript, Node.js, MySQL, NLP (TF-IDF, Cosine Similarity)*

- Built a **plagiarism detection engine** using TF-IDF and cosine similarity to identify and flag semantically similar submissions
- Designed **pairwise similarity analysis pipeline** for scalable comparison across student assignments
- Integrated **ML-driven integrity checks** into a role-based academic evaluation system
- Developed normalized **MySQL schemas** for efficient storage of submissions, similarity scores, and evaluation data

Car Catalogue System – Web Application | *HTML, CSS, JavaScript*

- Built a **responsive web app** for showcasing car brands, models, and specs with dynamic UI components
- Improved usability with intuitive layout and responsive design principles; designed for future scalability

RESEARCH WORK

Hypervolume in Higher Dimensions (4D Geometry) — *Research Paper (In Progress)*

- Investigating **hypervolume mathematical frameworks** in 4D space using linear algebra and higher-dimensional calculus
- Nearing completion with plans for final review and potential publication

Cardinality in Higher-Dimensional Sets — *Research Paper (In Progress)*

- Extending **cardinality concepts** from discrete mathematics to multidimensional contexts with formal mathematical analysis

SKILLS

Languages: C, C++, Python

Frameworks/Libraries: Streamlit, scikit-learn, RDKit, Pandas, NumPy

Tools: Git, VS Code, GitHub

Concepts: Machine Learning